Q 1

a) $\overline{(\bar{A}B + \bar{B}C) + \overline{(D + \bar{B}C)}}$

$= \overline{(\bar{A}B + \bar{B}C) + (\bar{D} \cdot \overline{(\bar{B} \cdot C)})}$

$= \overline{(\bar{A}B + \bar{B}C) + (\bar{D} \cdot (B + \bar{C}))}$

$= \overline{(\bar{A}B + \bar{B}C) + (B\bar{D} + \bar{C}\bar{D})}$

$= \overline{(\bar{A}B + \bar{B}C)} \cdot \overline{(B\bar{D} + \bar{C}\bar{D})}$

$= (\overline{\bar{A}B} \cdot \overline{\bar{B}C}) \cdot (\overline{B\bar{D}} \cdot \overline{\bar{C}\bar{D}})$

$= ((A + \bar{B}) \cdot (B + \bar{C}))$

$$((\overline{B}+D)\cdot(C+D))$$

$$=(AB+A\overline{C}+\overline{B}\,\overline{C})$$

$$(\overline{B}C+\overline{B}D+CD+D)$$

$$=(AB+A\overline{C}+\overline{B}\,\overline{C})$$

$$(\overline{B}C+D(\overline{B}+C+1)$$

$$=(AB+A\overline{C}+\overline{B}\,\overline{C})(\overline{B}C+D)$$

$$=AB.\overline{B}C+ABD+A\overline{C}.\overline{B}C$$

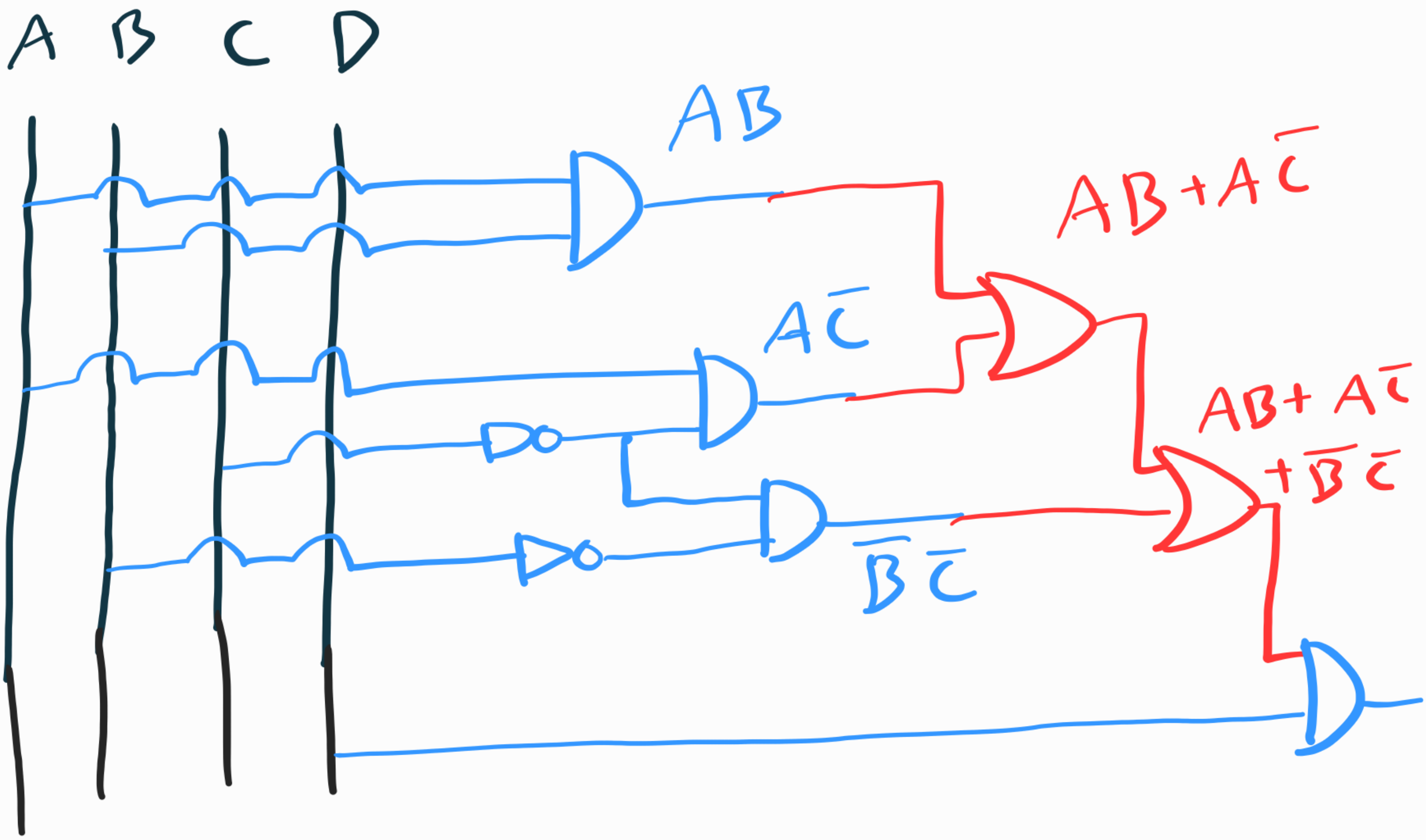
$$+\;A\overline{C}D+\overline{B}\,\overline{C}.\overline{B}C+$$

$$\overline{B}\,\overline{C}D$$

$$=ABD+A\overline{C}D+\overline{B}\,\overline{C}D$$

$$= D(AB + A\bar{C} + \bar{B}\,\bar{C})$$

(Ans)



b) $F(x, y, z) = \prod(1, 4, 6, 7)$

$$= (x + y + z')(x' + y + z)(x' + y' + z)(x' + y' + z')$$

$$= (xx' + xy + xz + x'y + y + yz + x'z'$$

$+ yz' + z'z)(x' + x'y' + x'z' +$
$x'y' + y' + y'z' + x'z + y'z + zz')$

$= (xy + xz + x'y + y + yz + x'z' +$
$yz') \cdot (x' + x'y' + x'z' + x'y' +$
$y' + y'z' + x'z' + y'z)$

$= (y(x + x' + 1 + z + z') + xz + x'z')$
$(x'(1 + y' + z' + y + z) +$
$y'(1 + z' + z))$

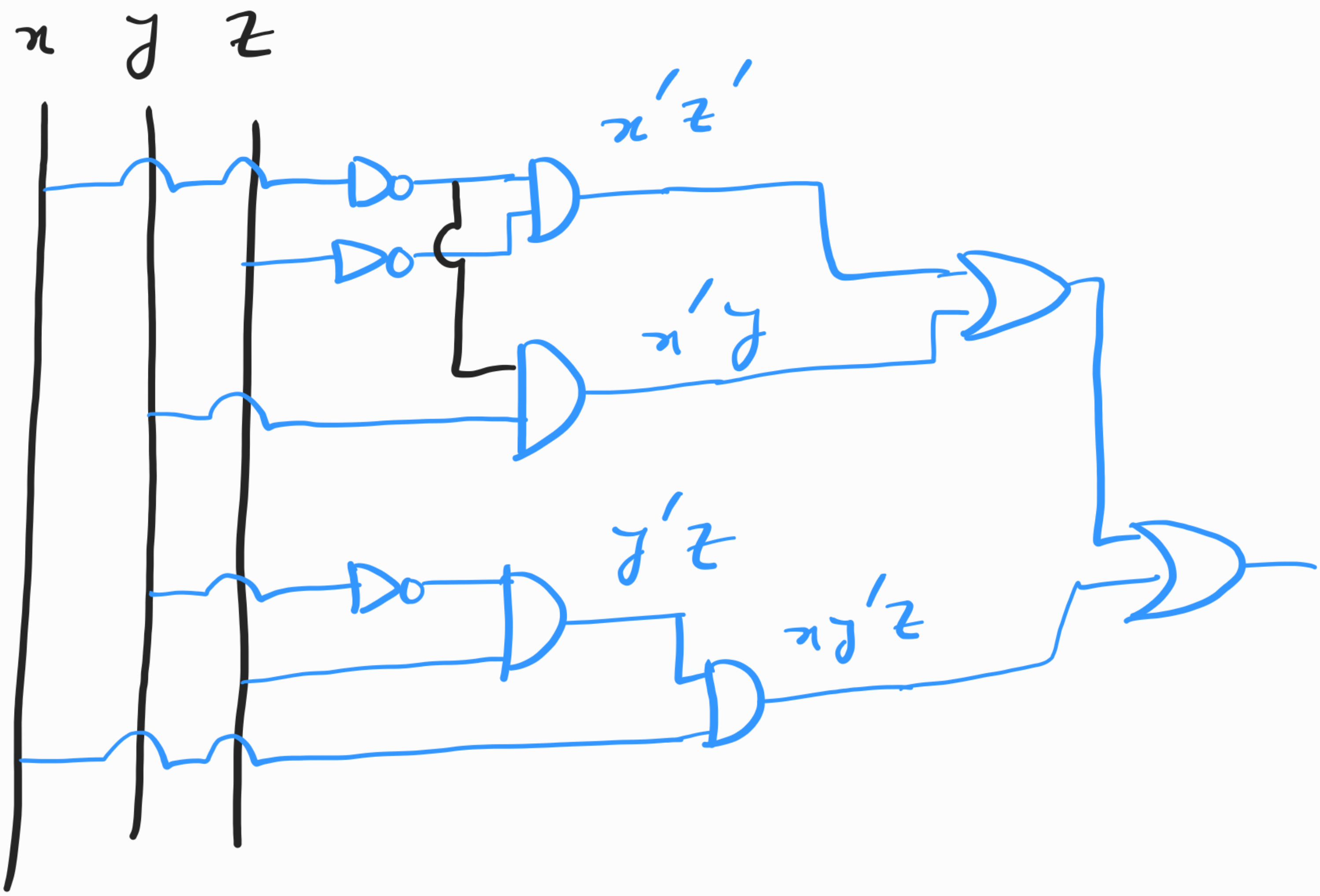
$= (y + xz + x'z')(x' + y')$

$= x'y + x'xz + x' \cdot x'z' + yy'$
$+ xy'z + x'y'z'$

$$= x'y + x'z' + xy'z + x'y'z'$$

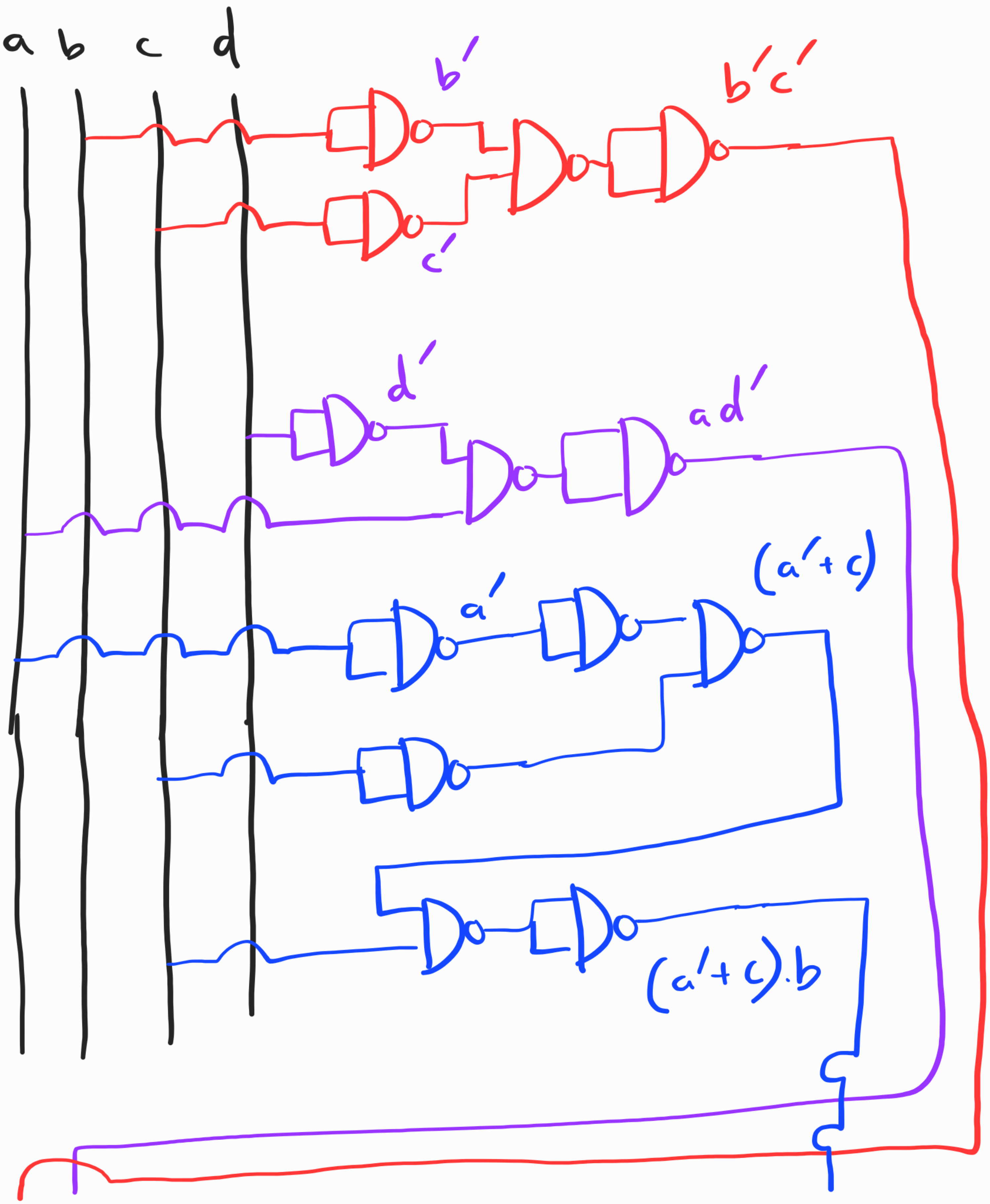
$$= x'z'(1 + y') + x'y + xy'z$$

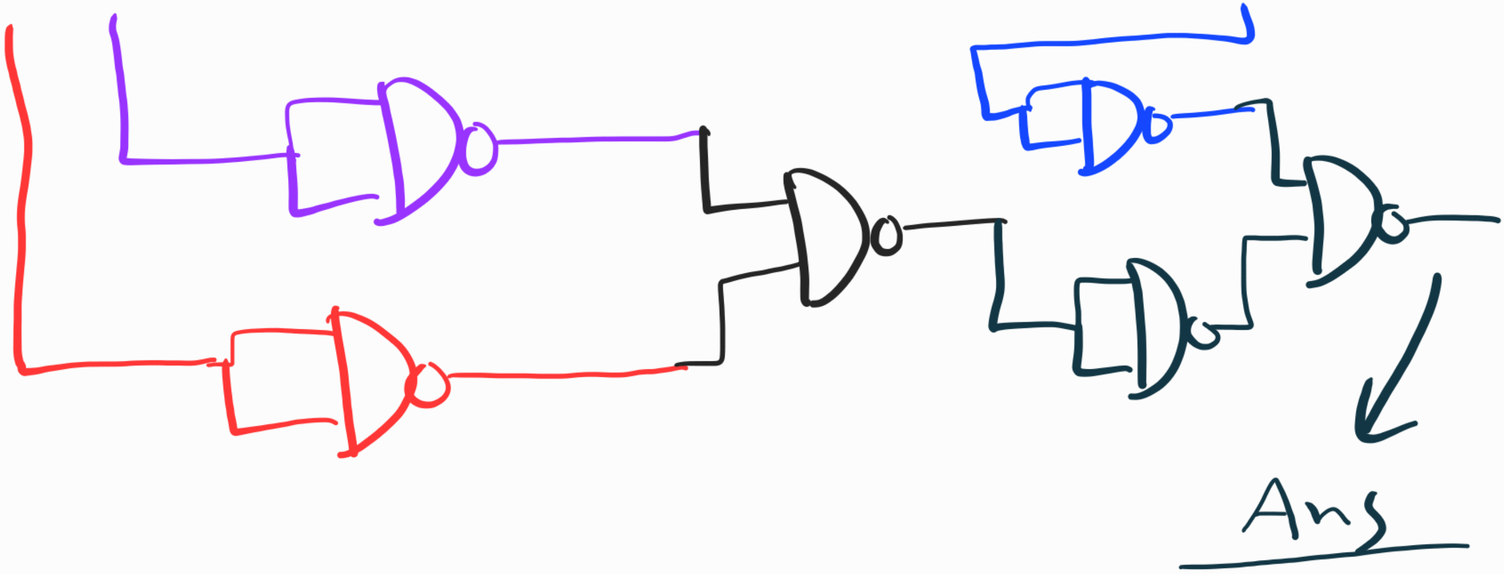
$$= x'z' + x'y + xy'z$$



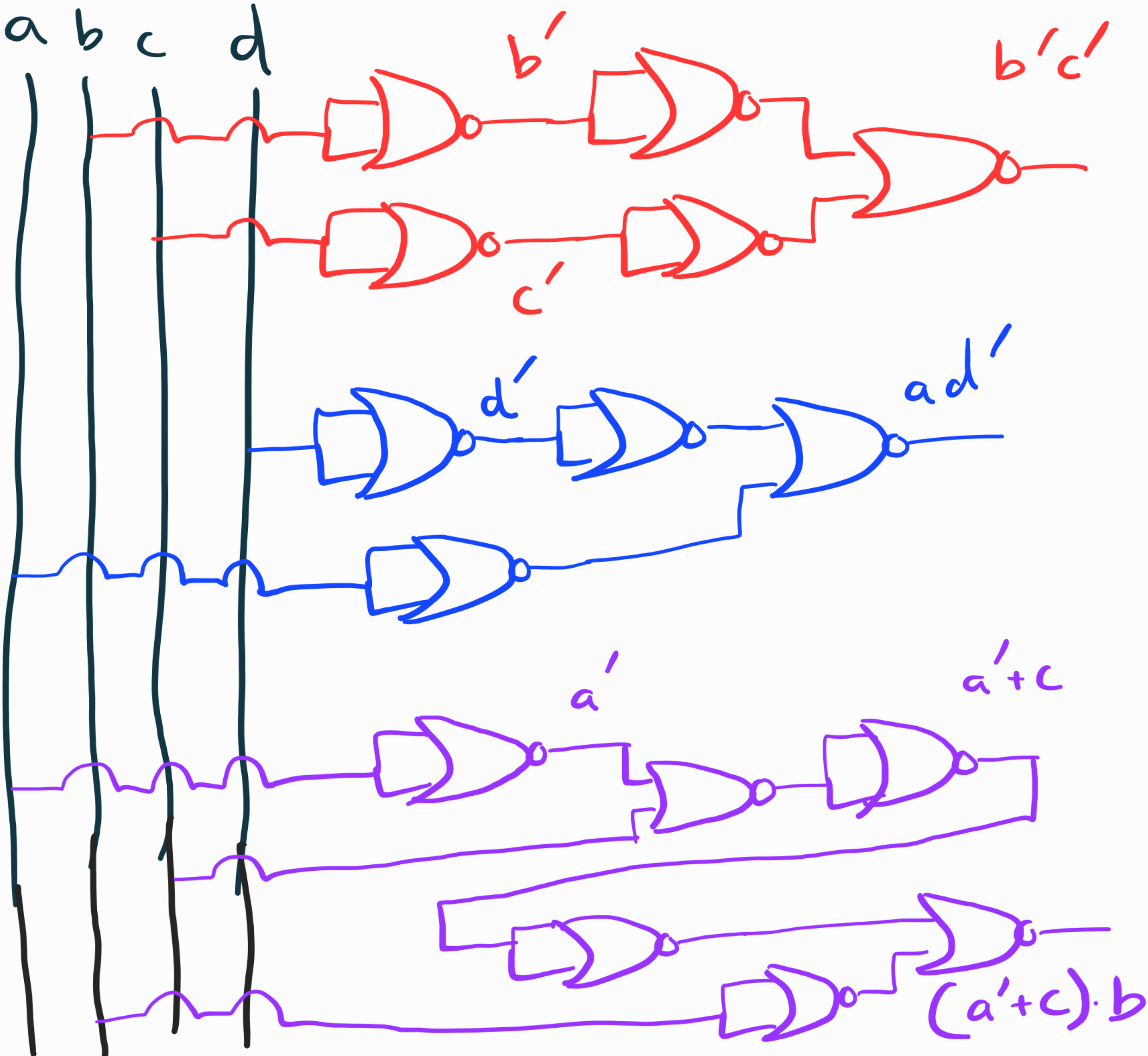
Q2 (NAND)

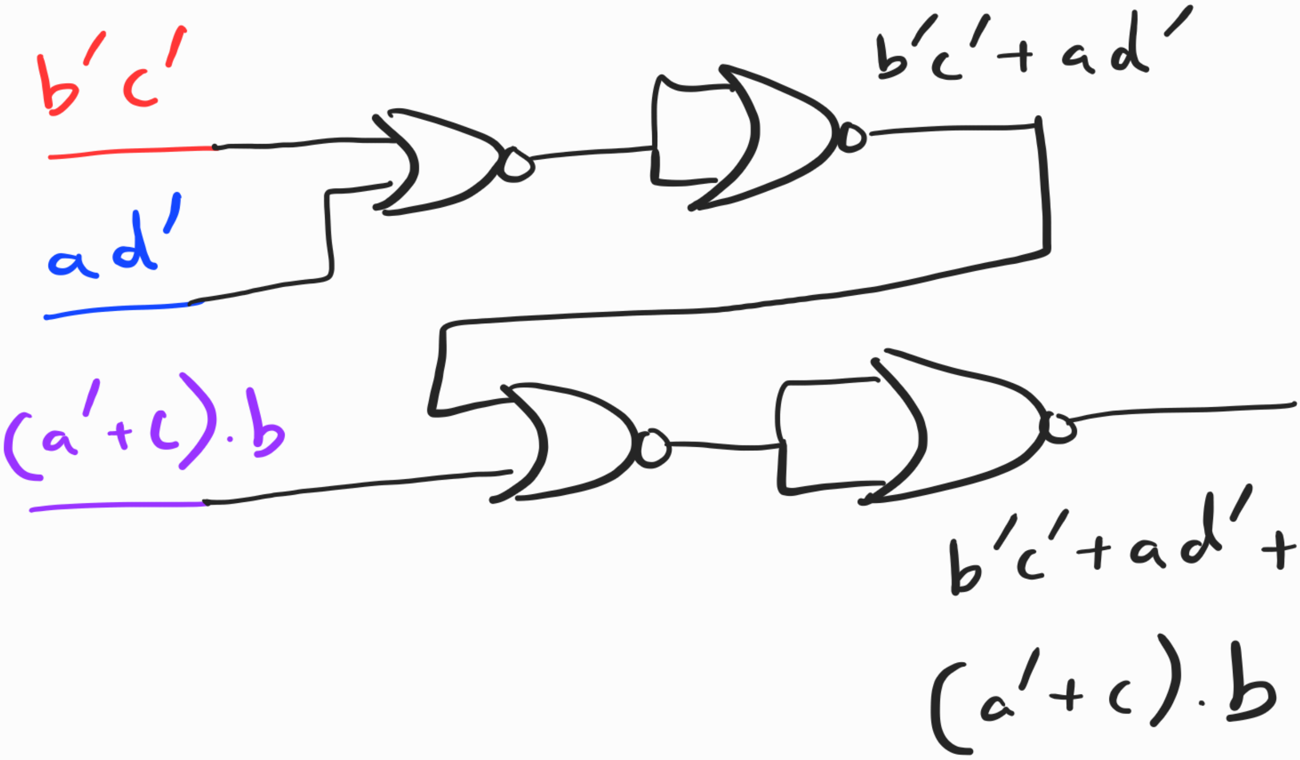
b) $b'c' + ad' + (a'+c)b$





NOR





Q3

b) $F(w,x,y,z) = \underbrace{x'y}_{x} + \underbrace{y}_{y}\underbrace{(z + x'z)}_{z}$

$= (\underbrace{x'y}_{yz} + \underbrace{y}_{x})(x'y + \underbrace{z}_{x} + \underbrace{x'z}_{yz})$

$= (y + x')(y + y)(x'y + (z + x')(z + z))$

$= (x' + y) \cdot y \cdot (\underbrace{x'y}_{x} + \underbrace{(x' + z)}_{y} \cdot \underbrace{z}_{z})$

$$= (x' + y + zz')(y + zz')$$

$$(\underbrace{x'y}_{yz} + \underbrace{z}_{x})(\underbrace{x'y}_{yz} + \underbrace{x' + z}_{x})$$

$$= (x' + y + z)(x' + y + z')(y + z)$$
$$(y + z')(x' + z)(y + z)$$
$$(x' + z + x')(x' + y + z)$$

$$= (x' + y + z)(x' + y + z')(y + z)$$
$$(y + z')(x' + z)$$

$$= (x' + y + z + ww')(x' + y + z' + ww')$$
$$(y + z + xx')(y + z' + xx')(x' + z + yy')$$

$$= (w + x' + y + z)(w' + x' + y + z)$$

$$(w + x' + y + z')(w' + x' + y + z')$$

$$(x + y + z)\underline{(x' + y + z)}$$

$$(x + y + z')(x' + y + z')$$

$$\underline{(x' + y + z)}(x' + y' + z)$$

$$= (w + x' + y + z)(w' + x' + y + z)$$

$$(w + x' + y + z')(w' + x' + y + z')$$

$$(x + y + z + ww')(x' + y + z + ww')$$

$$(x + y + z' + ww')(x' + y + z' + ww')$$

$$(x' + y' + z + ww')$$

$$= \underline{(w + x' + y + z)}\,\underline{(w' + x' + y + z)}$$

$$\underline{(w + x' + y + z')}\,\underline{(w' + x' + y + z')}$$

$(w+x+y+z)(w'+x+y+z)$

$\underline{(w+x'+y+z)}\ \underline{(w'+x'+y+z)}$

$(w+x+y+z')(w'+x+y+z')$

$\underline{(w+x'+y+z')}\ \underline{(w'+x'+y+z')}$

$(w+x'+y'+z)(w'+x'+y'+z)$

$= 0100, 1100, 0101, 1101,$
$0000, 1000, 0001, 1001,$
$0110, 1110$

$= \prod(4, 12, 5, 13, 0, 8,$
$1, 9, 6, 14)$

$= \prod(0, 1, 4, 5, 6, 8, 9, 12, 13,$
$14)$

Q4

a) $a'bc' + a'b'$

<u>General Rule:</u>

Dual Form,

$(a' + b + c') \cdot (a' + b')$

compliment,

$(a'' + b' + c'')(a'' + b'')$

$= (a + b' + c)(a + b)$

<u>De Morgan's Rule:</u>

$(a'bc' + a'b')'$

$= (a'bc')'(a'b')'$

$$= (a'' + b' + c'')\,(a'' + b'')$$

$$= (a + b' + c)\,(a + b)$$

Q5

b) $F(P, Q, R, S) = Q' + RS' + QS$

$$= RS' + \underbrace{Q'}_{x} + \underbrace{QS}_{yz}$$

$$= RS' + (Q' + Q)(Q' + S)$$

$$= \underbrace{RS'}_{yz} + \underbrace{Q' + S}_{x}$$

$$= (Q' + R + S)\underbrace{(Q' + S + S')}_{1}$$

$$= \underbrace{Q' + R + S}_{x} + \underbrace{PP'}_{yz}$$

$$= \underbrace{(P + Q' + R + S)(P' + Q' + R + S)}_{\text{Canonical POS}}$$

$= 0100,\ 1100$

$= \prod(4, 12)$ (maxterms)

$= \sum(0, 1, 2, 3, 5, 6, 7, 8, 9, 10,$
$11, 13, 14, 15)$ (minterms)